

31338 Network Servers

Lab 4a: Configuring a DHCP server on CentOS

Aims

1. Configure the CentOS as a DHCP server
2. Test the configuration by using a Windows client

Background Information

This lab will use VMnet2 (private network) to create a LAN shared by CentOS and Windows Server.

We will set up CentOS as a DHCP server, with the Windows Server as a DHCP client. Then switch the roles of each other.

Task 1: Install the Kea Package

Run the following command in the terminal

```
dnf install kea
```

Task 2: Configure and Start DHCP Server on CentOS

1. Use the same subnet based on Lab 3a. This lab will use the 10.0.2.0/24 subnet with the default route going to the interface *ens192*. This interface will be allocated a static IP address 10.0.2.1.
2. Read the *kea-dhcp4.conf* file at: */etc/kea/kea-dhcp4.conf* and then edit it. You can make a copy of it to the same directory as a backup.
3. The DHCP parameters are in the *option-data* section:
 - set the same domain-name-servers (DNS) with the local setting
 - set the same domain-search with the local setting
 - set the subnet
 - set the pool range, e.g., 10.0.2.100 – 10.0.2.200
 - set the same router with the local setting
 - set the reservation IP, bind with MAC address (optional). Note the MAC address format is with colons between each pair of hex digits.
4. Check the syntax using

```
kea-dhcp4 -t kea-dhcp4.conf
```

5. CentOS disables the service installed from other packages in the firewall by default. You need to enable the kea-dhcp service and reload the firewall

```
firewall-cmd --add-service=dhcp --permanent
```

```
firewall-cmd --reload
```

6. Use *systemctl* to start and verify the kea-dhcp4 service. No need to enable it to start at boot.

Task 3: Use Windows as the DHCP Client

1. Set up the Windows Server as a DHCP
(Hint: Just set the IP to automatic.)
2. Refresh the network settings in PowerShell. Sometimes you should disable the network interface *ens160* to refresh the settings.
3. You should see an IP address in the correct range and DNS and gateway settings correctly set
4. On the CentOS DHCP server, check the lease log at */var/lib/kea/kea-leases4.csv*.

Do not forget to record the steps in your learning journal.